

Exercise 2 - Theoretical question #1: solvent effects

How the solvent was included in Exercise 1

The water calculations in Exercise 1 used the conductor-like polarizable continuum model, CPCM(Water). This is an implicit model: there are no water molecules in the calculation. Instead, the molecule is placed in a cavity inside a uniform dielectric that has the bulk properties of water, $\epsilon = 80.4$ and refractive index $n = 1.33$.

The cavity is built from overlapping spheres on the atoms. ORCA 5 takes 1.2 times the van der Waals radius of each element (C 1.70, H 1.10, O 1.52, Cl 1.75 Å), which gives C 2.04, H 1.32, O 1.824 and Cl 2.1 Å for 2-chlorophenol. The cavity surface is divided into small elements, 755 Lebedev points in this case, and each element carries one charge.

These surface charges q_i represent the polarisation of the solvent by the solute. If the surrounding medium were a perfect conductor ($\epsilon \rightarrow \infty$), the total electrostatic potential on the surface would be zero. Written for all surface points this gives the linear equations $A q = -V$, where V_i is the potential of the solute (electrons and nuclei) at point i and A contains the Coulomb interactions between the surface charges. Water is not a conductor, so the charges are scaled by $f(\epsilon) = (\epsilon - 1)/(\epsilon + x)$: $A q = -f(\epsilon)V$. CPCM uses $x = 0$ ($f = 0.9876$ for water), while the original COSMO model uses $x = 0.5$.

In ORCA 5 each surface charge is a small Gaussian rather than a point charge, $A_{ii} = \zeta_i (2/\pi)^{1/2}/F_i$ and $A_{ij} = \text{erf}(\zeta_{ij} r_{ij})/r_{ij}$, where the switching function F_i gradually removes a charge when it moves into a neighbouring sphere. With point charges, surface points would appear and disappear as the atoms move and the energy would jump. The Gaussian charges keep the energy and its derivatives smooth, which is needed for the analytic gradients and Hessians used in the Opt and Freq runs.

The surface charges and the electron density depend on each other, so they are solved together in the SCF (self-consistent reaction field). In each iteration the potential of the charges, $V_{\text{solv}}(r) = \sum_i q_i / |r - r_i|$, is added to the one-electron part of the Kohn–Sham operator, the new density gives new charges, and this is repeated until both are converged. The quantity minimised is the free energy $G = \langle \Psi | \hat{H}^0 | \Psi \rangle + \frac{1}{2} \langle \Psi | \hat{V} | \Psi \rangle$, where the factor $\frac{1}{2}$ accounts for the work needed to polarise the solvent. All properties in Exercise 1 (orbitals, charges, dipole moment, frequencies) therefore belong to the polarised molecule in solution, which explains the larger dipole moment and the larger O–H charge separation in water.

The “CPCM Dielectric” energy in the output is the electrostatic part of the solvation free energy: -0.00746 Eh ($-4.68 \text{ kcal mol}^{-1}$) with PBE and -0.00763 Eh ($-4.79 \text{ kcal mol}^{-1}$) with B3LYP. Plain CPCM in ORCA 5 does not include the non-electrostatic terms (cavity formation, dispersion and repulsion), as the output also states.

Other ways to treat solvent effects in ORCA

SMD (Marenich, Cramer and Truhlar) uses the same CPCM electrostatics with its own atomic radii and adds a non-electrostatic term for cavitation, dispersion and solvent structure, calculated from

atomic surface tensions and solvent parameters. It is fitted to experimental solvation free energies, so it is the better choice when ΔG_{solv} itself is needed (%cpcm smd true end in ORCA 5, ! SMD(solvent) in ORCA 6).

openCOSMO-RS (ORCA 6) starts from the screening charges of a conductor calculation and treats the contacts between surface segments of solute and solvent with statistical thermodynamics. It gives solvation free energies in pure solvents and mixtures, including their temperature dependence.

A continuum cannot describe specific interactions such as a hydrogen bond between a water molecule and the OH group of 2-chlorophenol. For this, a few explicit water molecules can be added around the solute, by hand or with the SOLVATOR tool in ORCA 6, and the whole cluster can then be placed in CPCM (cluster–continuum model).

For a larger explicit solvent shell, ORCA offers QM/MM, where the solute is treated quantum mechanically and the solvent with a force field. Molecular dynamics with explicit solvent, as in Exercise 6, samples many solvent configurations instead of a single averaged one.